

GCE A LEVEL MARKING SCHEME

PREDICTED PAPER – ADVANCED

A LEVEL

ECONOMICS – COMPONENT 1

A520U10-1

GENERAL MARKING GUIDANCE

Positive marking should be applied: credit what the learner writes, rather than penalising omissions. The indicative content is not exhaustive; learners do not need to cover all points to reach the top band. Level-based mark schemes sub-divide the total mark across assessment objectives (AO1 Knowledge & Understanding, AO2 Application, AO3 Analysis, AO4 Evaluation). Learners may achieve different bands within a single question.

SECTION A — ANSWERS

Q	Ans	Notes / Working
1	C	Solow model: labour and capital growth exhibit diminishing returns in steady state; only exogenous TFP (technological progress) sustains long-run growth in per-capita GDP.
2	B	A Pigouvian tax set at marginal external cost shifts MPC up to MSC; output falls to the socially optimal level where MSB = MSC. The externality is 'internalised' — pollution is not eliminated but reduced to the socially optimal quantity.
3	C	Under a natural monopoly LRAC is falling, so $MC < AC$ over the relevant range. Regulating $P = MC$ means $P < AC \rightarrow$ the firm makes a loss (requires subsidy or different regulatory rule, e.g. average-cost pricing).
4	B	At $P = £50$: $Q_d = 600 - 4(50) = 400$; $Q_s = -100 + 6(50) = 200$. Excess demand = $400 - 200 = 200$ units. Equilibrium P would be £70 (not enforced here).
5	B	Real ER change $\approx$ Nominal ER change + $(\pi_{\text{home}} - \pi_{\text{foreign}}) = +1\% + (6\% - 2\%) = +5\%$ appreciation. UK goods become relatively more expensive abroad.
6	B	LRPC is vertical because in the long run workers correctly anticipate inflation and build it into wage demands, so there is no lasting output/employment gain from higher inflation (Friedman–Phelps expectations-augmented Phillips curve).
7	B	Progressive redistribution narrows the gap between richest and poorest quintile shares, flattening the Lorenz curve towards the 45° line and lowering the Gini coefficient.
8	A	Pre-tax: $120 - 2Q = 20 + 3Q \rightarrow Q = 20$, $P = £80$; $CS = \frac{1}{2} \times (120 - 80) \times 20 = £400$. Post-tax (supply shifts up £10): $120 - 2Q = 30 + 3Q \rightarrow Q = 18$, $P_c = £84$; $CS = \frac{1}{2} \times (120 - 84) \times 18 = £324$. Loss of CS = $£400 - £324 = £76$.
9	B	Perfect contestability requires zero sunk costs and free entry/exit; even with few incumbents, the threat of 'hit-and-run' entry forces firms to price at normal profit in the long run (Baumol).
10	B	In listed firms, shareholders (principals) delegate decision-making to managers (agents). Asymmetric information and divergent objectives (profit vs empire-building, salary, perks) create agency costs.
11	B	At the ZLB, the policy rate is already at (or very close to) 0% and cannot be cut further to stimulate demand without unconventional tools (QE, forward guidance, negative rates in some jurisdictions).
12	C	In open economies, leakages to imports, partial crowding-out of private investment, and offsetting exchange-rate movements all reduce the effective multiplier below its theoretical closed-economy value (often 0.5–1.5 empirically).
13	C	With perfectly inelastic demand, consumers absorb the entire tax; quantity demanded is unaffected. Tax revenue = tax $\times$ Q is therefore maximised (no demand response).
14	B	TOT improvement = export prices rise relative to import prices. With volumes unchanged (Marshall-Lerner not binding here since Q doesn't

		respond), the value of exports rises relative to imports → current account improves.
15	D	QE works via asset-price, portfolio-rebalancing, exchange-rate and signalling channels. It does NOT guarantee increased bank lending to SMEs — banks may hold excess reserves rather than lend (a key critique of QE post-2008).
16	B	Behavioural economics (Thaler & Sunstein) shows consumers are not fully rational; small changes to defaults, framing, or choice architecture can produce large behavioural changes without restricting choice — the 'libertarian paternalist' approach.
17	D	Commons are rival (one person's use reduces availability for others) and non-excludable (cannot prevent access). This combination leads to over-exploitation: each user captures the full benefit but only a fraction of the cost.
18	C	Real GDP growth = (Nominal growth)/(Deflator growth) – 1 = (2184/2000) / (104/100) – 1 = 1.092/1.04 – 1 = 1.05 – 1 = 5.0%.
19	C	Kuznets (1955): inequality first rises as industrialisation draws labour from low-productivity agriculture into higher-productivity industry (raising dispersion), then falls as wages converge and redistributive policies emerge — an inverted-U shape.
20	C	Pre-tax eq: $500 - 10P = 20P - 100 \rightarrow P = £20, Q = 300$. With £6 tax on producers, effective supply: $Q_s = 20(P - 6) - 100 = 20P - 220$. New eq: $500 - 10P = 20P - 220 \rightarrow 30P = 720 \rightarrow P_c = £24, Q = 260$.

21.	Using the diagram, explain why price stability is a common feature of oligopoly markets.	Total
	AO1: 2 marks Award 2 marks for good knowledge of the kinked-demand model (Sweezy, 1939): rivals match price cuts but ignore price rises, producing a demand curve that is elastic above and inelastic below P^*, with a discontinuity in MR. Award 1 mark for limited understanding or incorrect use of the diagram. AO3: 1 mark Award 1 mark for analysis: firms' MC can shift within the MR 'gap' without altering profit-maximising output or price, so prices stay sticky at P^*. Indicative content: • If one firm RAISES price above P^*, rivals do not follow → the raising firm loses substantial market share (elastic upper demand segment) → TR falls → disincentive to raise price. • If one firm CUTS price below P^*, rivals match → the firm gains little market share (inelastic lower demand segment) → TR falls → disincentive to cut price. • The discontinuity in MR (the vertical gap A–B) means that for a range of MC values the profit-maximising condition $MC = MR$ still yields the same output Q^* and price P^* → price is 'sticky'. • This explains the empirical observation that oligopolists tend to compete through non-price means (advertising, product differentiation) rather than price wars.	3

22. Consider the extent to which a £50bn expansionary fiscal stimulus would be an effective policy response. Adapt the diagram as part of your answer. (6 marks)

Band	AO1 (2)	AO3 (2)	AO4 (2)
2	2 – Good knowledge Clear grasp of Keynesian AS (flat → upward-sloping → vertical). AD shifts right by $k \times £50\text{bn}$ due to multiplier. Diagram correctly shows $AD_1 \rightarrow AD_2$ along flat portion of AS, with ΔY but no ΔP . Fully credit any reference to the multiplier with MPC.	2 – Good analysis Strong reasoning: in deep recession, spare capacity means fiscal stimulus raises real GDP with minimal inflation. Multiplier effect strongest here ($k \approx 1/(1-MPC)$).	2 – Good evaluation Strong counter-argument: crowding-out risk (even in recession), time lags, deficit financing concerns, multiplier size uncertain empirically, Ricardian equivalence, political economy of reversal.
1	1 – Limited knowledge	1 – Limited analysis	1 – Limited evaluation
0	0	0	0

Indicative content:

- £50bn of government spending shifts AD to the right. Under the Keynesian multiplier $k = 1/(1 - MPC)$, if $MPC = 0.6$, total AD shift $\approx £125\text{bn}$.
- On the flat portion of Keynesian AS (deep recession, high unemployment, idle capacity), the shift raises real GDP from Y_1 to Y_2 with no increase in the price level — the Keynesian 'free lunch'.
- Diagram: $AD_1 \rightarrow AD_2$; along flat AS, ΔY is large, ΔP is zero. Only as Y approaches full employment does the price level start rising (as AS becomes upward-sloping/vertical).

For the view that fiscal stimulus is effective:

- In deep recession, monetary policy is often ineffective (ZLB, liquidity trap) → fiscal becomes the only credible lever (IMF research, 2010s).
- Public investment (infrastructure) can raise LRAS as well as AD, so the stimulus has supply-side benefits too (non-inflationary growth).
- Automatic stabilisers supplement discretionary policy.
- Historical examples: Roosevelt's New Deal; China's 2008–09 stimulus; US CARES Act (2020).

BUT (evaluation):

- Crowding out: even in recession, bond issuance may raise long-term interest rates and crowd out private investment (Ricardian equivalence in the extreme).
- Time lags: recognition, decision, implementation and effect lags mean stimulus may arrive as the economy is already recovering — pro-cyclical.
- Deficit financing worsens national debt; if debt-to-GDP becomes unsustainable, bond markets may punish the government (UK mini-budget 2022 as recent example).
- Empirical multiplier estimates vary widely (0.5 to 2.5); the true size is uncertain.
- Import leakages in open economies reduce domestic impact.
- Quality of spending matters — malinvestment (politically-driven projects) may not deliver multiplier effects.
- Allow any other sensible evaluation.

23.	With the aid of a diagram, comment on the likely consequences for lithium prices over the next 5 years, referring to the price elasticity of supply.	Total
	AO1: 2 marks Award 2 marks for good understanding of PES determinants (time period, stock availability, spare capacity, factor mobility). Accurate diagram showing a steep (inelastic) short-run supply curve with a large rightward demand shift producing a large ΔP and small ΔQ. Award 1 mark for limited understanding; diagram missing or inaccurate. AO2: 1 mark Award 1 mark for using the stimulus (7–10 year lead times, existing mines at capacity, 4× demand) effectively to justify very inelastic short-run PES and the resulting sharp price rise. Indicative content: • PES is very low in the short run because mine expansion takes 7–10 years — the time period is shorter than the adjustment window, and existing capacity is nearly exhausted. • Diagram: near-vertical short-run supply curve. $D_1 \rightarrow D_2$ (demand quadruples): price rises sharply from P_1 to a much higher P_2, with a relatively small increase in quantity. • Consequences: battery-input cost inflation; pressure on EV manufacturers' margins; incentive for substitution (sodium-ion, solid-state batteries); strategic stockpiling; possible export restrictions by producing nations. • In the long run, PES becomes more elastic as new mines come online and recycling scales up, so prices may ease — cobweb-type dynamics possible.	3

24.	Explain the shape of this firm's short-run total cost curve and the economic significance of the discontinuity.	Total
	AO1: 2 marks Award 2 marks for good understanding of fixed vs variable costs, returns to a factor, and the significance of 'lumpy' capital investment producing a discontinuous (step) cost function. Award 1 mark for limited understanding with gaps or confusion. AO3: 2 marks Award 2 marks for strong chain of reasoning linking each segment of the TC curve to underlying economic concepts (division of labour, specialisation, integer problem of capital). Award 1 mark for underdeveloped analysis. Indicative content: • 1–99 units (artisan stage): TC rises from fixed cost base. Increasing returns to skilled labour produce a gently-rising TC curve. At $Q=99$, $TC = £5,000$. • Jump at $Q=100$ from £5,000 to £25,000 (£20,000 step): reflects the indivisible capital investment required to shift to factory production — machinery, premises, tooling. Classic 'lumpy' fixed cost. • 100–500+ units (factory stage): TC rises gradually again, initially at a shallower gradient than the artisan stage (economies of scale — machinery spreads fixed cost over higher output), but eventually rising as diminishing returns set in. • Economic significance: (i) the firm should only invest in the factory stage if the expected output justifies the £20,000 extra fixed cost; (ii) there is a break-even output above which factory production is cheaper per unit than artisan; (iii) the discontinuity explains why some firms hover at $Q=99$ despite unmet demand — the jump in TC makes the next unit uneconomic at low output.	4

25. With reference to the matrix and economic theory, discuss whether the optimal trade policy is unilateral free trade. (8 marks)

Band	AO2 (3)	AO3 (2)	AO4 (3)
3	3 – Excellent application Matrix fully used. Both countries' strategies analysed. Dominant strategies identified; uses specific growth rates (+2.4, +3.0, -3.5, -4.0). Notes asymmetry (X developed, Y emerging).		3 – Excellent evaluation Strong counter-argument using optimum tariff theory, strategic trade policy, infant industry, Rodrik's 'trilemma', or WTO context. Recognises unilateral free trade has theoretical appeal but political economy constraints.
2	2 – Good application Matrix used; both strategies discussed with some reference to the growth rates.	2 – Good analysis Strong reasoning on comparative advantage / gains from trade / welfare gains from unilateral liberalisation.	2 – Good evaluation Strong counter-argument about when unilateral free trade may not be optimal. Theory use more limited.
1	1 – Limited application	1 – Limited analysis	1 – Limited evaluation
0	0	0	0

The grid assumes that the main line is in favour of unilateral free trade, but the answer is fully reversible.

Indicative content:

AO2 (matrix interpretation):

- The (Free trade, Free trade) outcome gives the highest joint gains: (+2.4, +3.0). Total world welfare is maximised here.
- The (Trade war, Trade war) outcome is the worst jointly: (-3.5, -4.0). Mutual destruction.
- Note the asymmetry: Economy X (large developed) loses 3.5% in a trade war; Economy Y (emerging) loses 4.0% — Y is more exposed because a larger share of its GDP is trade-dependent.
- If country X engages unilaterally in free trade while Y protects: X gets some welfare gains from lower import prices and consumer surplus, but loses on export market access. Depending on the pay-offs in the off-diagonal cells (not shown), unilateral free trade may still be better than a trade war.
- Free trade is the Pareto-optimal outcome; trade war is the Nash equilibrium if one-shot (each fears defection). Repeated play enables free trade through reputation.

AO3 (if in favour of unilateral free trade):

- Consumer welfare: unilateral tariff removal lowers prices, widens choice, raises CS — the classic argument (Smith, Ricardo).
- Comparative advantage: resource reallocation to most-efficient sectors raises productivity and real GDP.
- Competitive pressure on domestic firms forces efficiency gains (dynamic/X-efficiency).
- Lower input costs for producers → lower export prices → potentially higher export competitiveness.
- Historical cases: UK's 19th-century Corn Law repeal; Hong Kong's free-port model; Chile's 1970s unilateral liberalisation.

AO4 (reasons unilateral free trade may NOT be optimal):

- Optimum tariff theory: a large country can improve its TOT by levying a tariff (the tariff revenue + TOT gain can exceed DWL if foreign PES is inelastic).

- Strategic trade policy (Brander-Spencer): targeted support for industries with increasing returns can shift rents to domestic firms.
- Infant industry argument (List, Hamilton): protection allows developing-country firms to achieve MES before competing.
- Employment/transition costs: displaced workers in uncompetitive sectors may not re-skill quickly; the China Shock literature (Autor et al.) documents durable labour-market damage.
- Strategic/defence industries require protection regardless of comparative advantage.
- Dumping and unfair subsidies by trading partners make unilateral openness vulnerable to predatory pricing.
- Political economy: losers concentrate, winners dispersed → democratic pressure favours protection (Olson).
- Rodrik's globalisation trilemma: cannot have hyperglobalisation, democracy and national sovereignty simultaneously.
- Revenue consideration for developing countries: tariff revenue is easy to collect and funds infrastructure (columns 5 and 6 of the matrix reflect this via Y's higher WTO bound tariff).
- Allow any other sensible evaluation.

26. With reference to the data above, consider the extent to which the UK's carbon pricing policies are effective at correcting the market failure of climate change. (8 marks)

Band	AO2 (3)	AO3 (2)	AO4 (3)
3	3 – Excellent application Quantitative data (48% CO ₂ fall, 43% renewables, £40 vs £240 carbon price, PED –0.3/–0.7) and textual data used together to build argument.		3 – Excellent evaluation Strong counter-argument using government failure, regulatory capture, carbon leakage, distributional concerns, global collective-action problem (1% UK share).
2	2 – Good application Numerical and textual data used to support arguments.	2 – Good analysis Strong reasoning on how a Pigouvian-style carbon price internalises the negative externality (Pigou/Coase/Stern Review).	2 – Good evaluation Counter-argument that UK carbon pricing may not effectively correct the failure; theory use more limited.
1	1 – Limited application	1 – Limited analysis	1 – Limited evaluation
0	0	0	0

The grid assumes that the main line is in favour of carbon pricing effectiveness, but the answer is fully reversible.

Indicative content:

AO2:

- UK emissions down 48% since 1990 — significant decoupling of emissions from growth; renewables share up from 7% to 43% (a sixfold rise), consistent with carbon pricing working.
- BUT the ETS price of £40/tonne is far below the Climate Change Committee's recommended shadow price of £240/tonne by 2030 — suggesting the externality is only partially priced.
- Fuel duty frozen since 2011 — in real terms, the carbon price on road transport has fallen, weakening the behavioural signal. Combined with short-run petrol PED of only –0.3, fuel-duty effectiveness is limited.
- ETS revenue fell from £5.5bn (2023) to £3.2bn (2024) — carbon price volatility reduces policy predictability and firm-level investment confidence.
- UK is only ~1% of global emissions; domestic effectiveness limited if major emitters (China, India, US) do not follow.

AO3 (if effective):

- Classic Pigouvian logic: externality of CO₂ → socially optimal output is where MSB = MSC, achieved by a tax or tradeable permit equal to the marginal external cost. Internalises the externality.
- Cap-and-trade (ETS) has the advantage of quantity certainty: the cap guarantees emissions reductions regardless of the price.
- Carbon pricing raises cost of carbon-intensive activities → substitution to renewables/efficiency → confirmed by 48% decline in emissions and 43% renewables share.
- Dynamic efficiency: persistent carbon price signal incentivises R&D into clean technologies (induced innovation).
- Long-run PED for petrol (–0.7) is more elastic than short-run (–0.3): price signal effective over time as consumers adjust vehicles, housing, behaviour.

AO4 (reasons carbon pricing may NOT effectively correct the failure):

- Government failure: the price (£40) is set too low versus the shadow cost (£240), leaving most of the externality unpriced — MSC still exceeds MPC for most CO₂ emissions.
- Only ~30% of emissions are covered by the UK ETS; large sectors (transport, buildings, agriculture) face weaker or no carbon signal.
- Carbon leakage: if UK firms face high carbon prices, production relocates to low-carbon-price jurisdictions — emissions not reduced, only outsourced. UK emissions on a consumption basis have fallen much less than on a production basis.
- Regressive distributional impact: carbon prices affect low-income households proportionately more (home heating, petrol are necessities) → political opposition (cf. French 'gilets jaunes', 2018).
- Frozen fuel duty shows political economy constraints — governments reluctant to raise visible costs.
- Climate change is a global public good / collective-action problem: UK's 1% share means unilateral pricing has limited direct impact on global temperatures. Free-riding is the dominant strategy.
- Regulatory/command approaches (boiler bans, EV mandates) may deliver more certain reductions than price signals alone.
- Volatility of ETS price (£90 → £40 in one year) undermines long-term investment signals.
- Allow any other sensible evaluation.

27. Using the information above and relevant economic theory, discuss the extent to which institutional quality is the most fundamental determinant of long-run economic development. (8 marks)

Band	AO2 (3)	AO3 (2)	AO4 (3)
3	3 – Excellent application All columns used. Clear comparisons — e.g., South vs North Korea (same ethnicity/geography, opposite institutions); Chile vs Argentina; Botswana vs Zimbabwe. Rule-of-law index integrated.		3 – Excellent evaluation Strong counter-argument: geography (Diamond, Sachs), culture (Weber), human capital (Lucas), luck/initial conditions. Engages with causality/reverse-causality problem.
2	2 – Good application Data used to support the argument; at least one strong natural-experiment comparison.	2 – Good analysis Strong chain: inclusive institutions → property rights → investment → growth (Acemoglu-Robinson).	2 – Good evaluation Alternative determinants of development considered. Theory use more limited.
1	1 – Limited application	1 – Limited analysis	1 – Limited evaluation
0	0	0	0

The grid assumes that the main line is in favour of the institutional-quality hypothesis, but the answer is fully reversible.

Indicative content:

AO2 (data interpretation):

- The Korean natural experiment is decisive: South Korea (HDI $\approx$ 0.93, inclusive institutions, high rule-of-law index) vs North Korea (HDI $\approx$ 0.5, extractive institutions). Same ethnicity, geography, initial conditions — divergent institutions produced radically different development outcomes.
- Chile (HDI $\approx$ 0.86, stronger rule of law) vs Argentina (HDI $\approx$ 0.84, weaker, prone to populist cycles): similar geography, different policy/institutional trajectories produced a growing gap from the 1970s.
- Botswana (HDI $\approx$ 0.73, strong institutions post-1966) vs Zimbabwe (HDI $\approx$ 0.55, catastrophic institutional collapse under Mugabe): both landlocked, similar natural resource bases — divergent institutions explain the gap.
- Singapore (HDI $\approx$ 0.94, world-leading rule of law despite being soft-authoritarian) vs Nigeria (HDI $\approx$ 0.54, oil-rich but institutionally fragile): oil is not a substitute for governance.
- The correlation between rule-of-law index and HDI in the data is strong, consistent with the institutional-quality hypothesis.

AO3 (if institutional quality IS the fundamental determinant):

- Acemoglu & Robinson (Why Nations Fail, 2012): inclusive institutions (broad property rights, inclusive political voice, rule of law, constraints on executive power) generate investment, innovation and growth. Extractive institutions (elite capture, weak property rights) produce stagnation.
- Property rights secure returns to investment (human and physical capital). Without them, the incentive to accumulate is weak (North, 1990).
- Rule of law reduces transaction costs, expands the reach of markets, and disciplines corruption (De Soto).
- Inclusive institutions are self-reinforcing: wider participation produces political accountability, which protects institutions from capture.

- Empirical: Acemoglu-Johnson-Robinson (2001) use settler-mortality as an instrumental variable, finding a causal effect of institutions on current GDP per capita.
- The 'resource curse' is institutionally conditioned: Norway and Botswana handle resource wealth well; Venezuela and Nigeria do not — difference is institutional.

AO4 (reasons institutions may NOT be the most fundamental determinant):

- Geography hypothesis (Diamond, Sachs): tropical disease burden, landlocked status, and soil quality strongly predict development. Africa's challenges partly reflect malaria, tsetse fly, and remoteness from trade routes.
- Culture/religion (Weber's Protestant work ethic; Putnam's social capital; Greif's cultural beliefs): values of thrift, trust, and cooperation may be deep drivers of which institutions emerge.
- Human capital (Lucas, 1988; Glaeser et al., 2004): some argue causality runs from education → institutions rather than the reverse. Educated populations demand better institutions.
- Reverse causality: rich countries can AFFORD better institutions — so the correlation may not be causal in the direction institutionalists claim.
- Initial conditions and path dependence matter — lucky accidents of history (the Industrial Revolution in Britain) shape trajectories.
- Dependency theory (Prebisch-Singer, Frank): historical exploitation of the 'periphery' by the 'core' constrains institutional development. The colonial inheritance of extractive institutions (as A&R themselves note) is a geographic/historical imposition, not just an institutional choice.
- Washington Consensus critique: institutional reforms imposed externally often failed (e.g. 1990s Russia privatisation without rule of law); institutions must be home-grown, which qualifies policy implications.
- China's 1978–present growth under formally extractive (one-party) institutions challenges the strict inclusive-institution thesis — suggests high-quality 'growth-promoting' institutions can take many forms.
- In practice, institutions, geography, human capital, culture and luck all interact — isolating 'the most fundamental' is methodologically contested.
- Allow any other sensible evaluation.

ASSESSMENT OBJECTIVE BREAKDOWN

Question	AO1	AO2	AO3	AO4	Total	QS
1–20	6	14	–	–	20	14
21	2	–	1	–	3	2
22	2	–	2	2	6	2
23	2	1	–	–	3	1
24	2	–	2	–	4	2
25	–	3	2	3	8	4
26	–	3	2	3	8	4
27	–	3	2	3	8	4
Total	14	24	11	11	60	33

A520U10-1 EDUQAS GCE A level Economics – Component 1 MS (Predicted, Advanced tier)